# SUMS OF TWO SQUARES WITH FEW PRIME FACTORS VIA THETA SERIES OF QUATERNARY FORMS

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ABSTRACT. Let $N \equiv 2 \pmod 4$ be large. We show that the number of $n \le N$ such that $n$ and $N-n$ are both sums of two squares, $\Omega(n(N-n)) \le 9$ and $\max(\Omega(n), \Omega(N-n)) \le 6$ is $\gg N/(\log N)^2$. The previous bounds, obtained by sifting the primes $\equiv 3 \pmod 4$ and switching, were 11 and 8. The new input is a level of distribution $2/5$ for the weighted sequence $r(n)r(N-n)$, where $r$ counts representations as a sum of two squares: for coprime $m, m'$ the sum $\sum_{ma+m'b=N} r(a)r(b)$ is the $N$-th coefficient of the weight-two theta series of $m(x^2+y^2)+m'(z^2+w^2)$, and Waibel's uniform bound for the Petersson norm of the cuspidal part of a theta series, combined with Deligne's bound through the Hecke operators, gives $\sum_{ma+m'b=N} r(a)r(b) = (\text{Siegel main term}) + O(d(N)N^{1/2}(mm')^{1/4+\varepsilon})$ uniformly in $m, m'$. Numerically the error does not grow with $mm'$ at all, which suggests that the level of distribution is in fact $1/2$.

## 1. INTRODUCTION

Let $r(n) = \#\{(x,y) \in \mathbb{Z}^2 : \ x^2+y^2 = n\}$, so that $r(n) = 4\sum_{d|n} \chi(d)$ with $\chi = \chi_{-4}$ the non-trivial character modulo 4. Every integer $N$ is a sum $n + (N-n)$ of two sums of two squares, and in [3] we asked for such representations in which $n$ and $N-n$ have few prime factors. For $N \equiv 2 \pmod 4$ it was shown there, by a vector sieve in which the primes $q \equiv 3 \pmod 4$ are sifted up to nearly $N^{1/2}$ and the survivors with two large such primes are removed by a switching argument, that there are $\gg N/(\log N)^2$ integers $n \le N$ with $n$, $N-n$ sums of two squares and $\Omega(n(N-n)) \le 11$, and $\gg N/(\log N)^2$ with $\max(\Omega(n), \Omega(N-n)) \le 8$.

Here we take a different route: instead of sifting the primes $\equiv 3 \pmod 4$ we count with the weight $r(n)r(N-n)$, which vanishes unless both $n$ and $N-n$ are sums of two squares. The price is that the sequence $r(n)r(N-n)$ must be shown to be well distributed in arithmetic progressions, and the gain is that no inert prime has to be sifted and no switching is needed. We prove a level of distribution $2/5$ (Theorem 3.2), and the sieve then gives the following.

**Theorem 1.1.** *Let $N \equiv 2 \pmod 4$ be sufficiently large. Then*

$$\#\big\{n \le N : \ n = x^2+y^2, \ N-n = z^2+w^2, \ \Omega(n(N-n)) \le 9, \ \max(\Omega(n), \Omega(N-n)) \le 6\big\} \ge \ c_1 \, \mathfrak{S}_r(N) \, \frac{N}{(\log N)^2},$$

*where $c_1 > 0$ is an absolute constant and*

$$\mathfrak{S}_r(N) = \prod_{\substack{p|N \\ p>2}} \rho_p, \qquad \rho_p = \begin{cases} \dfrac{p(p+1)(p-1)^2}{(p^2-p+1)^2} & p \equiv 1 \pmod 4, \\ \dfrac{p+1}{p}\Big(1 - \dfrac{1}{p(p-1)}\Big)^{-2} & p \equiv 3 \pmod 4. \end{cases} \tag{1}$$

*One has $\rho_p > 1$ for every $p$, so $\mathfrak{S}_r(N) \ge 1$. The integers counted are coprime to $N$, and each of $n$, $N-n$ is a product of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{9/250}$.*

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The constant can be made explicit, $c_1 = 0.25$ being admissible (Section 4); we have not tried to optimise it. Both bounds 9 and 6 come from one set of sieve parameters, so they hold simultaneously for the same integers $n$.

**The method.** Write a square-free $d$ as a product of primes $p \equiv 1 \pmod 4$ (split) and $q \equiv 3 \pmod 4$ (inert). The identities $r(n) = 2r(n/p) - r(n/p^2)$ for split $p \mid n$ and $r(n) = r(n/q^2)$ for inert $q \mid n$ express the sum of $r(n)r(N-n)$ over $n \equiv 0\ (d)$, $n \equiv N\ (e)$ as a combination of the sums

$$T(m,m';N) = \sum_{ma+m'b=N} r(a)r(b) = \#\{(x,y,z,w) \in \mathbb{Z}^4 :\ m(x^2+y^2)+m'(z^2+w^2) = N\} \tag{2}$$

with $m \mid d^2$, $m' \mid e^2$. This is the number of representations of $N$ by the positive quaternary form $Q_{m,m'} = m(x^2+y^2)+m'(z^2+w^2)$, i.e. the $N$-th Fourier coefficient of the theta series $\theta^2(mz)\theta^2(m'z)$, a holomorphic modular form of weight 2 on $\Gamma_0(4mm')$. By Siegel's theorem it is the sum of the $N$-th coefficient of an Eisenstein series, which is the product $M(m,m';N)$ of the local densities, and of the $N$-th coefficient of a cusp form $G_{m,m'}$. Deligne's bound controls the latter for each fixed form, and the whole difficulty is the uniformity in the level $4mm'$. This is exactly what Waibel's recent theorem on the Petersson norm of the cuspidal part of a theta series provides [12]: for our forms it gives $\|G_{m,m'}\| \ll (mm')^{1/4+\varepsilon}$. Passing from the norm to the $N$-th coefficient through the Hecke operator $T_N$ rather than through the Petersson formula directly avoids a loss for small levels, and we obtain

$$T(m,m';N) = M(m,m';N) + O_\varepsilon\big(d(N)^4 N^{1/2}(mm')^{1/4+\varepsilon}\big) \tag{3}$$

uniformly in coprime odd cube-free $m, m'$ (Theorem 2.3 and Corollary 2.6; for $(N, mm') = 1$ the factor $d(N)^4$ can be replaced by $d(N)$). Since the main term is of size $N/(mm')$, the error is smaller than the main term as long as $mm' \le N^{2/5-\varepsilon}$, and a short bookkeeping argument turns (3) into the level of distribution $2/5$ for $r(n)r(N-n)$ (Theorem 3.2).

The weighted sequence has the pleasant feature that under the weight $r(n)r(N-n)$ a split prime $p$ divides $n$ with density $h(p) = (2p-1)/(p(p+1)) \approx 2/p$ and an inert prime only with density $1/(q(q-1))$; so sifting all primes below $N^\alpha$ on both sides is a vector sieve of two linear components, and Richert's weights bound $\Omega(n(N-n))$. The numerical inequality behind Theorem 1.1 is verified with interval arithmetic (Section 4); it is considerably simpler than the one in [3], because there is no switching term.

**Comparison with the shifted convolution literature.** The sum (2) is a shifted convolution problem for the coefficients of $\theta^2$ with linear coefficients $m, m'$, in the "sum" orientation $ma+m'b = N$. For the divisor function in place of $r$ and the "difference" orientation, Topacogullari [11] proved an asymptotic formula uniform in the coefficients with error $(r_2x_1)^{1/2+\vartheta+\varepsilon}$, where $\vartheta = 7/64$ is the Kim–Sarnak exponent; the analogue of his result for $r(n)$ in our orientation would give the level $1/2 - \vartheta = 25/64$. The holomorphic structure of our problem does better in two ways: the sum orientation makes $T(m,m';N)$ a coefficient of a holomorphic form, so that Deligne's theorem replaces the Ramanujan–Petersson bound at finite places, and the uniformity in the level is supplied by the Petersson norm. The exponent $1/4$ in (3) is the price of a Cauchy–Schwarz step in the passage from the norm of $G_{m,m'}$ to a single coefficient. Numerically (Section 5) the quantity $\sup_N |T-M|/\sqrt{N}$ does not grow with $mm'$, while the root mean square decreases like $(mm')^{-1/4}$, which shows that Waibel's bound is sharp up to $N^\varepsilon$ but suggests that the error in (3) is really $O(N^{1/2+\varepsilon})$ uniformly. If so, the level of distribution is $1/2$ and the bounds in Theorem 1.1 improve to 8 and 5 (Remark 4.5).

**Notation.** $p$ denotes a prime $\equiv 1 \pmod 4$ and $q$ a prime $\equiv 3 \pmod 4$ when the distinction matters; $\chi = \chi_{-4}$; $d(n)$ and $\omega(n)$, $\Omega(n)$ are the usual divisor and prime-factor counting functions; $\sigma_{-1}(n) = \sum_{t|n} 1/t$. The Petersson inner product on $S_2(\Gamma_0(L))$ is

$$\langle f, g\rangle = \int_{\Gamma_0(L)\backslash\mathbb{H}} f(z)\overline{g(z)}\,\mathrm{d}x\,\mathrm{d}y, \qquad \|f\|^2 = \langle f, f\rangle, \tag{4}$$

which is [6, (2.1)] and the normalisation of [12] for weight 2. All implied constants may depend on $\varepsilon$.

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## 2. Theta series of the forms $m(x^2+y^2)+m'(z^2+w^2)$

Throughout this section $m, m'$ are coprime odd positive integers, $Q = Q_{m,m'}$ is the quaternary form $m(x^2+y^2)+m'(z^2+w^2)$ with Gram matrix $A = \mathrm{diag}(2m, 2m, 2m', 2m')$, and

$$\Theta(z) = \theta^2(mz)\theta^2(m'z) = \sum_{n\ge 0} T(m, m'; n)e(nz), \qquad \theta(z) = \sum_{k\in\mathbb{Z}} e(k^2 z).$$

Since $\theta^2 \in M_1(\Gamma_0(4), \chi)$, we have $\Theta \in M_2(\Gamma_0(4mm'))$ with trivial character. In the terminology of [12], $Q$ is primitive (because $(m, m') = 1$), $\det A = 16(mm')^2$, and the level of $Q$, the least $L$ such that $LA^{-1}$ is integral with even diagonal, is $L = 4mm'$.

### 2.1. Siegel's theorem and the local densities.

By Siegel's theorem [10] (for weight 2 see Kitaoka [9] and [12, Section 2]), $\Theta = E + G$ where $E$ is the theta series of the genus of $Q$ (a linear combination of Eisenstein series) and $G \in S_2(\Gamma_0(4mm'))$ is a cusp form, and the coefficients of $E$ are given by the product of local densities [12, (1)–(2)]

$$M(m, m'; n) = \frac{(2\pi)^2 n}{\Gamma(2)\sqrt{\det A}} \prod_p \beta_p(n) = \frac{\pi^2 n}{mm'} \prod_p \beta_p(n), \tag{5}$$

where $\beta_p(n) = \lim_{a\to\infty} p^{-3a}\#\{x \in (\mathbb{Z}/p^a)^4 :\ Q(x) \equiv n\ (p^a)\}$. We record the densities that we need. For $p \nmid 2mm'$ the form is equivalent over $\mathbb{Z}_p$ to $x^2+y^2+z^2+w^2$ and

$$\beta_p(n) = (1-p^{-2})\,\sigma_{-1}(p^{v_p(n)}), \tag{6}$$

as follows from Jacobi's formula $r_4(n) = 8\sum_{t|n,\,4\nmid t} t$ (or from a direct count). For $p \mid mm'$ with $p \nmid n$ we have

$$\beta_p(n) = 1 - \frac{\chi(p)}{p}\,: \tag{7}$$

indeed, if $p \mid m$ then modulo $p$ the equation reads $m'(z^2+w^2) \equiv n$, which has $p - \chi(p)$ solutions $(z, w)$, none of them with $z \equiv w \equiv 0$; every such solution is non-singular and lifts to $p^{3(a-1)}$ solutions modulo $p^a$ by Hensel's lemma, so $\beta_p(n) = p^2(p-\chi(p))/p^3$. At $p = 2$ the density $\beta_2(n)$ depends only on $n$ modulo a bounded power of 2 and on whether $m, m'$ are $\equiv 1$ or $3 \pmod 4$; we shall only use that $\beta_2(n) \asymp 1$ for $v_2(n) \le 1$ and that for $m = m' = 1$ one has $M(1,1;n) = r_4(n)$ exactly (there are no cusp forms of level 4). In particular, for $(n, mm') = 1$,

$$\frac{M(m, m'; n)}{M(1,1;n)} = \frac{1}{mm'} \prod_{p|mm'} \frac{1-\chi(p)/p}{1-p^{-2}} = \frac{1}{mm'} \prod_{p|mm'} g(p), \tag{8}$$

where $g(p) = p/(p+1)$ for $p \equiv 1 \pmod 4$ and $g(q) = q/(q-1)$ for $q \equiv 3 \pmod 4$.

2.2. **Waibel's bound.**

**Lemma 2.1.** $\|G\| \ll (mm')^{1/4+\varepsilon}$.

*Proof.* Theorem 1 of [12] states, for a primitive integral positive quaternary form of level $L$ and $f = \theta(Q) - \theta(\mathrm{gen}Q)$,

$$\langle f, f\rangle \ll \frac{L^{2+\varepsilon}}{F(Q,2)} + \frac{L^{1+\varepsilon}}{(\det A)^{1/4}},$$

with the inner product (4) and a genus invariant $F(Q,s)$ defined in [12, (6)] from the Jordan decompositions of $Q$ over $\mathbb{Z}_p$: $F(Q,s) = 2^{\min(4-2r_2,s)}\prod_{p|L} p^{\mu_p(s)}$, $\mu_p(s) = \sum_{j=1}^{v_p(L)} \min(v_p(j), s)$, where $v_p(j)$ is the number of variables whose $p$-adic coefficient has valuation $\geq j$ and $r_2$ the number of binary 2-adic blocks. For $Q_{m,m'}$ and an odd prime $p$ with $p^a \parallel mm'$ the $p$-adic valuations of the coefficients are $(a, a, 0, 0)$, so $v_p(j) = 2$ for $j \leq a$ and $\mu_p(2) = 2a$; at $p = 2$ all coefficients are units, $r_2 = 0$ and $\mu_2(2) = 0$. Hence $F(Q,2) = 4(mm')^2$ (and $F(Q,4) = 16(mm')^2 = \det A$, as it must). With $L = 4mm'$ and $\det A = 16(mm')^2$ the bound becomes $\langle G, G\rangle \ll (mm')^{\varepsilon} + (mm')^{1/2+\varepsilon}$. □

2.3. **From the norm to a single coefficient.** The Petersson formula gives, for an orthonormal basis $\mathcal{B}$ of $S_2(\Gamma_0(L))$ and $n \geq 1$ (see [6, Prop. 2.1 and Cor. 2.2], which are stated for all even $k \geq 2$; we write them for the unnormalised coefficients $a_f(n)$),

$$\sum_{f\in\mathcal{B}} |a_f(n)|^2 = 4\pi n\,\Delta_L(n,n), \qquad \Delta_L(n,n) = 1 + O\Big(\frac{d(L)\,d_3(n)\,(n,L)^{1/2}\,n}{L\,(n+L)^{1/2}}\log 2n\Big) \tag{9}$$

with an absolute implied constant. Applying this directly at $n = N$ would cost a factor $(1 + N^{1/4}L^{-1/2})$ for small levels (this is [5, Cor. 14.24], used in [12, (3)]), which is too expensive for us; we route the coprime part of $N$ through a Hecke operator instead.

**Lemma 2.2.** *Let $L \geq 1$, $G \in S_2(\Gamma_0(L))$, and $n = n_1n_2$ with $(n_1, L) = 1$, $n_2 \mid L^{\infty}$ and $n_2 \leq L^{1/2}$. Then*

$$|a_G(n)| \leq C\,d(n_1)\,n^{1/2}\,\|G\|$$

*with an absolute constant $C$.*

*Proof.* For $(n_1, L) = 1$ the Hecke operator $T_{n_1}$ is self-adjoint on $S_2(\Gamma_0(L))$ with respect to (4) and is diagonalised by a basis of forms $f(tz)$, $f$ a newform of some level $L' \mid L$ and $t \mid L/L'$, with eigenvalue $a_f(n_1)$ ([8, Thm. 6.20], [1]). By Deligne's theorem $|a_f(n_1)| \leq d(n_1)n_1^{1/2}$, hence $\|T_{n_1}G\| \leq d(n_1)n_1^{1/2}\|G\|$. The $n_2$-th coefficient of $H = T_{n_1}G$ is $\sum_{t|(n_1,n_2)} t\,a_G(n_1n_2/t^2) = a_G(n)$ because $(n_1, n_2) = 1$. Finally, by Cauchy–Schwarz over an orthonormal basis and (9),

$$|a_H(n_2)|^2 \leq \|H\|^2 \sum_{f\in\mathcal{B}} |a_f(n_2)|^2 = 4\pi n_2\Delta_L(n_2,n_2)\|H\|^2,$$

and for $n_2 \leq L^{1/2}$ the error term in (9) is $\ll d(L)\,d_3(n_2)\,n_2^{3/2}\log(2n_2)/L^{3/2} \ll d(L)d_3(n_2)L^{-3/4}\log(2L) \ll 1$, so $\Delta_L(n_2, n_2) \leq C'$. □

**Theorem 2.3.** *Let $m, m'$ be coprime odd positive integers and $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$. Then*

$$T(m, m'; N) = M(m, m'; N) + O_{\varepsilon}\big(d(N)\,N^{1/2}(mm')^{1/4+\varepsilon}\big).$$

*Proof.* $T - M = a_G(N)$ with $G \in S_2(\Gamma_0(4mm'))$. Apply Lemma 2.2 with $n_1$ the odd part of $N$ and $n_2 = 2^{v_2(N)} \mid 4mm'$, then Lemma 2.1. □

*Remark* 2.4. For fixed $m, m'$ the error term $O(d(N)N^{1/2})$ is Deligne's bound and cannot be improved. The point of Theorem 2.3 is the dependence on $mm'$; see Section 5 for what it seems to be in truth.

2.4. **Primes of $mm'$ dividing $N$.** The sieve also needs $T(m,m';N)$ when a prime of $mm'$ divides $N$. Such terms reduce exactly to the coprime case. Let $U_p$ act on $q$-expansions by $U_p\sum a(n)e(nz) = \sum a(pn)e(nz)$, so that $a_N(\Phi) = a_{N/p}(U_p\Phi)$ when $p \mid N$, and note that $U_p\big(\Phi(pz)\Psi(z)\big) = \Phi(z)\,(U_p\Psi)(z)$. The identities $r(pb) = 2r(b) - r(b/p)$ for split $p$ and $r(qb) = r(b/q)$ for inert $q$ (with $r(x) = 0$ for $x \notin \mathbb{Z}$) say that for $p \nmid m'$

$$U_p\,\theta^2(m'z) = 2\theta^2(m'z) - \theta^2(pm'z) \quad (p \equiv 1\ (4)), \qquad U_q\,\theta^2(m'z) = \theta^2(qm'z) \quad (q \equiv 3\ (4)). \tag{10}$$

**Lemma 2.5.** *Let $p$ be an odd prime with $p \mid N$ and $p \mid m$, $p \nmid m'$.*

(i) *If $p \equiv 1\ (4)$ and $m = pm_1$ with $p \nmid m_1$: $T(pm_1,m';N) = 2T(m_1,m';N/p) - T(m_1,pm';N/p)$.*
(ii) *If $p \equiv 1\ (4)$ and $m = p^2m_1$: $T(p^2m_1,m';N) = 2T(pm_1,m';N/p) - T(m_1,m';N/p^2)$, where the last term is $0$ if $p^2 \nmid N$.*
(iii) *If $p \equiv 3\ (4)$ and $m = pm_1$ with $p \nmid m_1$: $T(pm_1,m';N) = T(m_1,pm';N/p)$.*
(iv) *If $p \equiv 3\ (4)$ and $m = p^2m_1$: $T(p^2m_1,m';N) = T(m_1,m';N/p^2)$, which is $0$ if $p^2 \nmid N$.*

*The same holds with the roles of $m$ and $m'$ exchanged. Consequently, if $v_p(mm') \le 2$ for every odd prime $p$, then $T(m,m';N)$ is a linear combination with integer coefficients of terms $T(\tilde m,\tilde m';\tilde N)$ with coprime $\tilde m,\tilde m'$, $\tilde N \mid N$, $v_2(\tilde N) = v_2(N)$, $(\tilde m\tilde m',\tilde N) = 1$ and $\tilde m\tilde m' \le mm'$, the sum of the absolute values of the coefficients being at most $\prod_{p\mid(mm',N)} 6v_p(N) \le 6^{\omega(N)}d(N)$. Since the identities hold for the modular forms themselves, they hold for the Eisenstein parts $M$ and the cusp parts separately.*

*Proof.* In case (i), $\theta^2(mz) = \Phi(pz)$ with $\Phi = \theta^2(m_1\,\cdot)$, so

$$T(pm_1,m';N) = a_{N/p}\big(\theta^2(m_1z)\,U_p\theta^2(m'z)\big),$$

and (10) gives the claim; (iii) is the same with the second identity. In case (ii), $\theta^2(mz) = \Phi(pz)$ with $\Phi = \theta^2(pm_1\,\cdot)$, so $T = 2T(pm_1,m';N/p) - T(pm_1,pm';N/p)$, and $T(pm_1,pm';N') = T(m_1,m';N'/p)$ (or $0$ if $p \nmid N'$) because the form is $p$ times $Q_{m_1,m'}$; (iv) is similar. For the last assertion fix $p$ with $p \mid (mm',N)$ and iterate: when $v_p(\tilde m\tilde m') = 1$ we apply (i) or (iii), when it is $2$ we apply (ii) or (iv). On the right-hand sides the two coefficients are coprime again, $v_p(\tilde N)$ has decreased by at least one, and exactly one term still contains $p$ in its coefficients (with $v_p = 1$); the others are of the final form. The coefficient of the continuing term has absolute value at most $2$ (it is $\pm1$ times the previous one in (i), (iii) and at most twice the initial one in (ii)), so the finished terms have coefficients of absolute value at most $4$. After at most $v_p(N)$ steps either $p \nmid \tilde N$ or $p \nmid \tilde m\tilde m'$, and the sum of the absolute values of all coefficients produced is at most $4v_p(N) + 2 \le 6v_p(N)$. Treating the primes of $(mm',N)$ one after the other gives the bound $\prod_p 6v_p(N) \le 6^{\omega(N)}d(N)$. The Eisenstein and cuspidal parts of $\Theta$ are preserved by $U_p$ and by $z \mapsto pz$. □

**Corollary 2.6.** *For coprime odd $m,m'$ with $v_p(mm') \le 2$ for all $p$, and $N$ with $v_2(N) \le 1$,*

$$T(m,m';N) = M(m,m';N) + O_\varepsilon\big(6^{\omega(N)}d(N)^2\,N^{1/2}(mm')^{1/4+\varepsilon}\big),$$

*and $6^{\omega(N)}d(N)^2 \le d(N)^4$.*

## 3. The level of distribution of $r(n)r(N-n)$

Let $N \equiv 2 \pmod 4$ and

$$\mathcal{A} = \{n :\ 1 \le n \le N,\ (n,N) = 1\}, \qquad w(n) = r(n)r(N-n).$$

Elements of $\mathcal{A}$ are odd, and $w(n) \neq 0$ forces $n \equiv N - n \equiv 1 \pmod 4$. For coprime square-free $d, e$ with $(de, 2N) = 1$ put

$$S(d,e) = \sum_{\substack{n \in \mathcal{A} \\ d|n,\ e|N-n}} w(n).$$

3.1. **Decomposition into theta coefficients.** For split $p$ and inert $q$ the relations $r(n) = 2r(n/p) - r(n/p^2)$ $(p \mid n)$ and $r(n) = r(n/q^2)$ $(q \mid n)$ give, for square-free $d$,

$$\sum_{d|n} r(n) e(nz) = \sum_{m|d^2} c_d(m)\, \theta^2(mz), \tag{11}$$

where $c_d$ is multiplicative in the sense $c_d(m) = \prod_{p|d} c_p(p^{v_p(m)})$ with $c_p(p) = 2$, $c_p(p^2) = -1$ for split $p$ and $c_q(q^2) = 1$, $c_q(q) = 0$ for inert $q$. Every $m$ with $c_d(m) \neq 0$ has the form $m = ds$ with $s \mid d$, and $\sum_m |c_d(m)| \leq 3^{\omega(d)}$. The condition $(n, N) = 1$ is removed by Möbius inversion over the square-free divisors $f$ of the odd part of $N$, and the parity of $n$ (equivalently of $a = n/fd$) by the identity $r(2a') = r(a')$: the terms with $a$ even in $T(m, m'; N)$ sum to $T(m, m'; N/2)$. Altogether

$$S(d,e) = \sum_{f|N^\flat} \mu(f) \sum_{m|d^2} \sum_{m'|e^2} c_d(m) c_e(m') \sum_{f_1|f^2} c_f(f_1)\, T^{\mathrm{odd}}(f_1 m, m'; N), \tag{12}$$

with $T^{\mathrm{odd}}(\cdot\,; N) = T(\cdot\,; N) - T(\cdot\,; N/2)$, where $N^\flat$ is the product of the odd primes dividing $N$. Each $T$ in (12) has coprime odd coefficients and is covered by Corollary 2.6.

3.2. **The main term.** Define $M(d,e)$ by replacing each $T$ in (12) by its Eisenstein part $M$. The Siegel densities are multiplicative over the primes, and the linear identities behind (11) and the Möbius inversion hold for the numbers of solutions modulo every prime power; hence $M(d,e)/M(1,1;N)-$ type ratios are products of conditional local densities. Concretely, let $X$ be the Eisenstein part of $S(1,1) = \sum_{n \in \mathcal{A}} w(n)$:

$$X = \big(r_4(N) - r_4(N/2)\big) \prod_{p|N^\flat} (1 - g_N(p)) = 16\, \sigma(N/2) \prod_{p|N^\flat} (1 - g_N(p)), \tag{13}$$

where we used that $M(1,1;n) = r_4(n)$, Jacobi's formula ($r_4(N) = 24\sigma(N/2)$, $r_4(N/2) = 8\sigma(N/2)$ for $N \equiv 2 \pmod 4$), and where $g_N(p)$ is the proportion of the $p$-adic solutions of $x^2+y^2+z^2+w^2 = N$ with $p \mid x^2 + y^2$. Then

$$M(d,e) = X\, h(d)\, h(e), \qquad h(p) = \frac{2p-1}{p(p+1)} \ \ (p \equiv 1\ (4)), \qquad h(q) = \frac{1}{q(q-1)} \ \ (q \equiv 3\ (4)), \tag{14}$$

$h$ multiplicative.

**Lemma 3.1.** *Let $p$ be odd. (i) If $p \nmid N$, the proportion of the $p$-adic solutions of $x^2+y^2+z^2+w^2 = N$ with $p \mid x^2 + y^2$ is $h(p)$. (ii) If $p^a \parallel N$, $a \geq 1$, this proportion is*

$$g_N(p) = 1 - \frac{\beta_p^*}{(1-p^{-2})\sigma_{-1}(p^a)}, \qquad \beta_p^* = \begin{cases} (1-1/p)^3 & p \equiv 1 \pmod 4, \\ (1-1/p)(1+1/p)^2 & p \equiv 3 \pmod 4; \end{cases}$$

*in particular $g_N(p) = 4p/(p+1)^2$ for $p \equiv 1 \pmod 4$, $p \parallel N$, and $g_N(q) = 0$ for $q \equiv 3 \pmod 4$, $q \parallel N$.*

*Proof.* (i) Modulo $p$, with $N$ a unit, the equation $x^2 + y^2 + z^2 + w^2 \equiv N$ has $p^3 - p$ solutions, all non-singular. Those with $x^2 + y^2 \equiv 0$ number $(2p-1)(p-1)$ if $p \equiv 1 \pmod 4$ ($2p-1$ choices of $(x,y)$ on the two lines $y \equiv \pm ix$, then $p-1$ choices of $(z,w)$ with $z^2 + w^2 \equiv N$) and $1 \cdot (q+1)$ if $q \equiv 3 \pmod 4$. Hensel's lemma lifts each solution to $p^{3(a-1)}$ solutions modulo $p^a$, so the proportion is $(2p-1)(p-1)/(p^3-p) = h(p)$, resp. $(q+1)/(q^3-q) = h(q)$. (ii) The solutions with

$p \nmid x^2+y^2$ are non-singular, and their number modulo $p$ is $\sum_{u\neq 0} \#\{x^2+y^2\equiv u\}\#\{z^2+w^2\equiv -u\} = (p-1)(p-\chi(p))^2$, which gives the density $\beta_p^*$; the total density is (6). □

Note that $(1-g_N(p))\sigma_{-1}(p^a) = \beta_p^*/(1-p^{-2})$ does not depend on $a$; this is why $\mathfrak{S}_r(N)$ in (1) has one factor per prime. The products in (13) and (14) follow from Lemma 3.1 by multiplicativity; for (14) one can also verify directly that $\sum_{m\mid d^2} c_d(m)\, g(\operatorname{rad} m)/m = h(d)$ with $g$ as in (8): for split $p$, $2g(p)/p - g(p)/p^2 = \frac{p}{p+1}\big(\frac{2}{p}-\frac{1}{p^2}\big) = h(p)$, and for inert $q$, $g(q)/q^2 = h(q)$.

3.3. **The remainder.** Put $R(d,e) = S(d,e) - M(d,e)$.

**Theorem 3.2.** *Let $N\equiv 2 \pmod 4$, $\varepsilon>0$ and $D = N^{2/5-\varepsilon}$. Then*

$$\sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\, d_3(de)\, |R(d,e)| \ \ll_\varepsilon\ N^{1-\varepsilon/3}.$$

*Proof.* Each $T$ in (12) has coefficients $(\tilde m,\tilde m') = (f_1 m, m')$ with $\tilde m\tilde m' = f_1\, ds\, es'$, $s\mid d$, $s'\mid e$, $f_1\le f^2$; they are coprime and cube-free because $d,e,f$ are pairwise coprime and square-free. Let $\eta=\varepsilon/10$. We use two bounds for $|T-M|$: Corollary 2.6,

$$|T(\tilde m,\tilde m';N') - M(\tilde m,\tilde m';N')| \ll d(N)^4\, N^{1/2}(\tilde m\tilde m')^{1/4+\eta} \qquad (N'\in\{N,N/2\}),$$

and the trivial bound $|T-M|\le T+M \ll N^{1+\eta}/(\tilde m\tilde m') + N^\eta$, valid because there are at most $N/(\tilde m\tilde m')+1$ pairs $(a,b)$ with $\tilde m a+\tilde m' b = N'$ and $r(a)r(b)\ll N^\eta$, while $M\ll N^{1+\eta}/(\tilde m\tilde m')$ by (5). All divisor-type factors ($d(N)^4$, $d_3(de)$, $3^{\omega(de)}$, $d(de)$, $d(N^\flat)^{O(1)}$) are $\ll N^\eta$. The terms with $f_1 ss' > N^\varepsilon$ are estimated trivially:

$$\sum_{de\le D} d_3(de)\, 3^{\omega(de)} \sum_{f\mid N^\flat} 3^{\omega(f)} \sum_{\substack{s\mid d,\ s'\mid e,\ f_1\mid f^2\\ f_1ss'>N^\varepsilon}} \Big(\frac{N^{1+\eta}}{f_1 ss' de} + N^\eta\Big)$$
$$\ll N^{1+2\eta-\varepsilon} \sum_{de\le D} \frac{d_3(de) 3^{\omega(de)} d(de)}{de} \sum_{f\mid N^\flat} \frac{9^{\omega(f)}}{f} + DN^{3\eta} \ll N^{1-\varepsilon+3\eta} + N^{2/5+3\eta},$$

and those with $f_1 ss' \le N^\varepsilon$ by the uniform bound:

$$\ll d(N)^4\, N^{1/2+\eta} \sum_{de\le D} d_3(de) 3^{\omega(de)} (de)^{1/4} \sum_{s\mid d,\, s'\mid e} \sum_{\substack{f\mid N^\flat,\ f_1\mid f^2\\ f_1ss'\le N^\varepsilon}} (f_1 ss')^{1/4}$$
$$\ll N^{1/2+\varepsilon/4+3\eta} D^{5/4} = N^{1-\varepsilon+3\eta}.$$

Since $3\eta = 3\varepsilon/10$, both bounds are $\ll N^{1-7\varepsilon/10} \le N^{1-\varepsilon/3}$. □

*Remark* 3.3. The exponent $2/5$ is $1/(2(1+\beta))$ with $\beta = 1/4$ the exponent of $mm'$ in Theorem 2.3; the terms with $m$ close to $d^2$, where the level $mm'$ is as large as $(de)^2$, are harmless because of the trivial bound. A uniform bound with $\beta=0$ (Conjecture 5.1) would give the level $1/2$.

## 4. The sieve

We keep the notation of Section 3. Fix $0<\alpha<\beta_1<2/5$ and $\lambda>0$, and put

$$z = N^\alpha, \qquad y_1 = N^{\beta_1}, \qquad P = \prod_{\substack{2<p<z\\ p\nmid N}} p.$$

For $n \in \mathcal{A}$ define Richert's weight

$$w_n = 1 - \lambda \sum_{\substack{p \mid n(N-n),\ p \equiv 1\,(4) \\ z \le p < y_1}} \Big(1 - \frac{\log p}{\log y_1}\Big), \qquad W = \sum_{\substack{n \in \mathcal{A} \\ (n(N-n),P)=1}} w_n\, w(n).$$

**Lemma 4.1** (Structure). *Let $n \in \mathcal{A}$ with $(n(N-n),P)=1$ and $w(n) \neq 0$. Then every prime factor of $n(N-n)$ is $\ge z$. Unless $n(N-n)$ is divisible by the square of a prime $\ge z$, both $n$ and $N-n$ are products of distinct primes $\equiv 1 \pmod 4$, each prime dividing exactly one of them, and $w(n) = 16 \cdot 2^{\Omega(n(N-n))}$. The total weight of the exceptional $n$ is $\ll N^{1-\alpha+\varepsilon}$.*

*Proof.* $(n,N)=1$ and $(n(N-n),P)=1$ leave only prime factors $\ge z$. An inert prime divides a sum of two squares only to an even power, so if $n(N-n)$ is square-free all its prime factors are split, and then $r(n) = 4 \cdot 2^{\omega(n)}$. For the exceptional $n$, $p^2 \mid n(N-n)$ for some $p \ge z$, and $\sum_{p \ge z} \sum_{p^2 \mid n(N-n)} w(n) \ll N^{\varepsilon} \sum_{p \ge z} N/p^2 \ll N^{1-\alpha+\varepsilon}$. □

**Lemma 4.2** (Prime factors). *If $n$ is as in Lemma 4.1, $n(N-n)$ square-free and $w_n > 0$, then $\Omega(n(N-n)) < 1/\lambda + 2/\beta_1$ and $\max(\Omega(n), \Omega(N-n)) < 1/\lambda + 1/\beta_1$.*

*Proof.* As in [3, Lemma 3.2]: write $w_n = 1 - \lambda(T_1 + T_2)$ with $T_1, T_2$ the sums over the primes dividing $n$, resp. $N-n$. Each term is non-negative, and since $n < N = y_1^{1/\beta_1}$ has fewer than $1/\beta_1$ prime factors $\ge y_1$, $T_1 \ge \Omega(n) - 1/\beta_1$; similarly for $T_2$. Then $w_n > 0$ gives $T_1 + T_2 < 1/\lambda$, hence both claims. □

**Proposition 4.3** (Main term). *Let $\theta = 2/5$, $\theta_A \in (0,\theta)$, $s_A = \theta_A/\alpha$, $s_B = (\theta - \theta_A)/\alpha$, and let $F_1, f_1$ be the upper and lower functions of the linear sieve. Put*

$$L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B), \qquad R = 2\int_{\alpha}^{\beta_1} \Big(1 - \frac{u}{\beta_1}\Big) U(\theta - u)\, \frac{\mathrm{d}u}{u},$$

*where $U(t) = \min_{t_A + t_B = t} F_1(t_A/\alpha)F_1(t_B/\alpha)$. Then*

$$W \;\ge\; (L - \lambda R + o(1))\, X\, V, \qquad V = \prod_{\substack{2 < p < z \\ p \nmid N}} (1 - h(p))^2,$$

*and*

$$XV = \big(1 + o(1)\big)\, 8\, \mathfrak{S}_r(N)\, c_h^2\, \frac{N}{\alpha^2 (\log N)^2}, \qquad c_h = \lim_{z \to \infty} (\log z) \prod_{2 < p < z} (1 - h(p)) = 1.50136\ldots \tag{15}$$

*Proof.* Apply the vector sieve inequality $ab \ge a^- b^+ + a^+ b^- - a^+ b^+$ [2] with $a = 1_{(n,P)=1}$, $b = 1_{(N-n,P)=1}$ and the Rosser–Iwaniec linear sieve weights [7] of levels $N^{\theta_A}$ and $N^{\theta - \theta_A - \varepsilon}$. By (14) the sequence $\{n \in \mathcal{A} :\ d \mid n\}$ with the weight $w$ has density $h(d)$, with $h(p) = (2p-1)/(p(p+1))$ on the split primes and $h(q) \ll q^{-2}$ on the inert ones; since $\sum_{p < z,\, p \equiv 1\,(4)} 1/p = \frac{1}{2} \log\log z + O(1)$, each side is a sieve of dimension 1, i.e. $\prod_{w \le p < z,\, p \nmid N} (1 - h(p))^{-1} \le (\log z / \log w)(1 + K/\log w)$ with an absolute $K$. The combined moduli are $\le N^{\theta - \varepsilon}$, and Theorem 3.2 shows that the remainders contribute $O(N^{1-\varepsilon/3}) = o(XV)$. This gives $\sum_{(n(N-n),P)=1} w(n) \ge (L + o(1))XV$. For the weights, the subsequence $\{n \in \mathcal{A} :\ p \mid n\}$ with $z \le p < y_1$, $p \nmid N$, has density $h(p)h(d)$ for $d \mid n$ and the same structure, and the upper bound vector sieve at total level $N^{\theta - u - \varepsilon}$, $p = N^u$, bounds its sifted weight by $(U(\theta - u) + o(1))h(p)XV$; the remainders are again controlled by Theorem 3.2 since $d_3(pde) \ge d(de)$ absorbs the sum over $p$. Since $\sum_{p \equiv 1\,(4),\ N^u \le p < N^{u + \mathrm{d}u}} h(p) = (1 + o(1))\, \mathrm{d}u/u$ by the prime number theorem in progressions ($h(p) = 2/p + O(p^{-2})$ on half of the primes), and

the side $N-n$ contributes the same amount, the weighted sum is bounded by $(R+o(1))XV$ with the factor 2 in $R$ accounting for the two sides. Finally, by (13), Lemma 3.1 and (1),

$$XV = 8N \prod_{p\mid N^\flat} \frac{\beta_p^*}{1-p^{-2}} \prod_{\substack{2<p<z\\ p\nmid N}} (1-h(p))^2 = 8N\,\mathfrak{S}_r(N) \prod_{2<p<z} (1-h(p))^2 \prod_{\substack{p\mid N\\ p\geq z}} (1-h(p))^2,$$

and the last product is $1+O(\omega(N)/z)$. The constant $c_h$ exists by Mertens' theorem in arithmetic progressions: $\prod_{2<p<z}(1-h(p)) = P_1(z)^2 \prod_{2<p<z}(1-h(p))(1-1/p)^{-2\cdot 1_{p\equiv 1(4)}}$ with $P_1(z) = \prod_{p<z,\,p\equiv 1(4)}(1-1/p) \sim (2e^{-\gamma}/\kappa)^{1/2}(\log z)^{-1/2}$, $\kappa = \frac{\pi}{4}\prod_{q\equiv 3(4)}(1-q^{-2})$, the last product converging absolutely. Numerically $\kappa = 0.672386\ldots$, the correction product is $0.898991\ldots$ and $c_h \in [1.50136, 1.50137]$ (script `singular_constant.py`, primes up to $10^7$ with explicit tail bounds). □

| $\alpha$ | $\lambda$ | $\beta_1$ | $1/\lambda+2/\beta_1$ | $1/\lambda+1/\beta_1$ | $L\geq$ | $R\leq$ | $L-\lambda R\geq$ |
|---|---|---|---|---|---|---|---|
| 9/250 | 3/10 | 3/10 | 10 | 20/3 | 0.998974 | 2.816690 | 0.153967 |
| 4/125 | 7/30 | 7/20 | 10 | 50/7 | 0.999637 | 3.471422 | 0.189639 |
| 9/250 | 1/3 | 3/10 | 29/3 | 19/3 | 0.998974 | 2.816690 | 0.060078 |

TABLE 1. Certified bounds at level $\theta = 2/5$ with $\theta_A = \theta_B = 1/5$. The first line gives $\Omega(n(N-n)) \leq 9$ and $\max(\Omega(n), \Omega(N-n)) \leq 6$; the second $\Omega(n(N-n)) \leq 9$ with the larger constant; the third the same bounds as the first with $\lambda = 1/3$.

### 4.1. **Numerical verification.**

**Lemma 4.4.** *The bounds of Table 1 hold.*

*Method of verification.* The computation is carried out by the script `certify_modular.py`, which follows [3, Lemma 4.1]. The functions $F_1, f_1$ are evaluated from their closed forms $F_1(s) = 2e^\gamma/s$ on $(0,3]$ and $f_1(s) = 2e^\gamma \log(s-1)/s$ on $[2,4]$ in interval arithmetic (`mpmath.iv`, 80 bits) and continued by the delay-differential equations $(sF_1)' = f_1(s-1)$, $(sf_1)' = F_1(s-1)$ on the grid $10^{-3}\mathbb{Z}$ with outward rounding, using that $F_1$ decreases and $f_1$ increases. $L$ is bounded below through $L = f_1(s_A)F_1(s_B) + F_1(s_A)(f_1(s_B) - F_1(s_B))$ with $f_1 - F_1 \leq 0$. $R$ is bounded above by a Riemann sum over 4000 cells: on a cell $[u_a, u_b]$ the factor $(1-u/\beta_1)/u$ is largest at $u_a$, the level $\theta - u$ is smallest at $u_b$, and $U$ is bounded by the product $F_1(t/\alpha)F_1((\theta-u_b-t)/\alpha)$ at one admissible $t$. The rational numbers $1/\lambda+2/\beta_1$ and $1/\lambda+1/\beta_1$ are computed exactly. □

*Proof of Theorem 1.1.* Take the first line of Table 1. By Proposition 4.3 and Lemma 4.4, $W \geq (0.153967+o(1))XV$. Since $w_n \leq 1$ and $w(n) \geq 0$, the sum of $w(n)$ over the $n$ with $(n(N-n), P) = 1$ and $w_n > 0$ is at least $W$; removing the exceptional $n$ of Lemma 4.1 costs $O(N^{1-\alpha+\varepsilon})$. By Lemmas 4.1 and 4.2 every remaining $n$ has $n$, $N-n$ sums of two squares, $\Omega(n(N-n)) \leq 9$, $\max(\Omega(n), \Omega(N-n)) \leq 6$ and $w(n) = 16\cdot 2^{\Omega(n(N-n))} \leq 2^{13}$. Hence the number of such $n$ is at least

$$\big(0.153967+o(1)\big)\,2^{-13}\cdot 8\,\mathfrak{S}_r(N)\,c_h^2\,\frac{N}{\alpha^2(\log N)^2} \;\geq\; 0.26\,\mathfrak{S}_r(N)\frac{N}{(\log N)^2}$$

for large $N$, since $0.153967\cdot 2^{-10}\cdot c_h^2\cdot(250/9)^2 \geq 0.2615$. Finally $\rho_p > 1$ for all odd $p$: for $p\equiv 1 \pmod 4$, $p(p+1)(p-1)^2 - (p^2-p+1)^2 = p^3-4p^2+3p-1 = p^2(p-4)+3p-1 > 0$ for $p\geq 5$ (e.g. $\rho_5 = 480/441$), and for $q\equiv 3 \pmod 4$, $\rho_q > (q+1)/q > 1$. □

*Remark* 4.5. If the error term in Theorem 2.3 were $O(N^{1/2+\varepsilon})$ uniformly in $m, m'$ (Conjecture 5.1), Theorem 3.2 would hold with $D = N^{1/2-\varepsilon}$, and the same computation at $\theta = 1/2$ with $\alpha = 6/125$, $\lambda = 1/3$, $\beta_1 = 17/50$ (so $1/\lambda + 2/\beta_1 = 151/17 < 9$, $1/\lambda + 1/\beta_1 = 101/17 < 6$) certifies $L \geq 0.997718$, $R \leq 2.498893$, $L - \lambda R \geq 0.164754$, hence $\Omega(n(N-n)) \leq 8$ and $\max(\Omega(n), \Omega(N-n)) \leq 5$ (script `certify_modular.py` with the argument `1/2`).

## 5. Numerical evidence and a conjecture

The quantities $T(m, m'; N)$ for all $N \leq X$ are obtained at once by a fast Fourier transform convolution of the sequences $r(a)1_{m|\cdot}$ and $r(b)1_{m'|\cdot}$, and $M(m, m'; N)$ from (5)–(7) with the 2-adic density computed exactly modulo $2^9$ (script `sup_error_scan.py`). Table 2 and Figure 1 report the statistics of $(T - M)/\sqrt{N}$ over all $N \in (X/2, X]$ with $(N, mm') = 1$ and $v_2(N) \leq 3$, for $X = 2^{21}$ and $X = 2^{23}$. Three consistency checks: for $mm' \in \{1, 3\}$ (levels 4 and 12, where $S_2(\Gamma_0(L)) = 0$) the error is identically zero, so the main term (5) is exact; for $(m, m') = (1, 5)$ and $(9, 1)$ the error vanishes identically for $N \equiv 2 \pmod 4$, because the newforms of levels 20 and 36 have $a(2) = 0$ and there are no oldforms with $a(2) \neq 0$; and the density (7) was checked against exact counts modulo $p^2$ for $p \leq 13$.

| $(m, m')$ | $mm'$ | rms | 99.9% quantile | sup, $X = 2^{21}$ | sup, $X = 2^{23}$ |
|---|---|---|---|---|---|
| $(1, 5)$ | 5 | 1.76 | 8.7 | 14.7 | 17.3 |
| $(9, 1)$ | 9 | 1.76 | 13.7 | 28.7 | 33.8 |
| $(3, 5)$ | 15 | 1.90 | 7.1 | 10.5 | 12.2 |
| $(1, 101)$ | 101 | 1.40 | 5.4 | 11.8 | 12.1 |
| $(13, 29)$ | 377 | 0.89 | 3.3 | 5.9 | 6.8 |
| $(1, 1009)$ | 1009 | 0.67 | 2.5 | 4.4 | 5.3 |
| $(61, 67)$ | 4087 | 0.38 | 1.4 | 2.5 | 2.4 |
| $(101, 103)$ | 10403 | 0.27 | 1.0 | 1.9 | 1.8 |
| $(1, 15015)$ | 15015 | 0.18 | 0.6 | 1.0 | – |
| $(1009, 1013)$ | 1022117 | 0.036 | 0.29 | 0.61 | 0.55 |

Table 2. Statistics of $(T(m, m'; N) - M(m, m'; N))/\sqrt{N}$ over all admissible $N \in (X/2, X]$ ($\approx 10^6$ values for $X = 2^{21}$, $4 \cdot 10^6$ for $X = 2^{23}$; the run with $X = 2^{23}$ covers a selection of the pairs).

Two features stand out. First, the root mean square decreases like $(mm')^{-1/4}$. By the Rankin–Selberg method, the mean square of $a_G(n)/\sqrt{n}$ over $n \leq Y$ tends to $(4\pi)^2\|G\|^2/\mathrm{vol}(\Gamma_0(4mm')\backslash\mathbb{H})$, and the volume is $\asymp mm'$; so the data say $\|G\| \asymp (mm')^{1/4}$, i.e. Lemma 2.1 is sharp up to $N^{\varepsilon}$. (Waibel's lower bound [12, Theorem 3] gives only $\|G\|^2 \gg 1/\min(m, m')$ for our forms.) Second, the supremum over $N$ does not grow with $mm'$ at all—it decreases—while Theorem 2.3 allows growth like $(mm')^{1/4}$. The gap is the Cauchy–Schwarz step in Lemma 2.2: the coefficient $a_G(N)$ is a sum over $\asymp mm'$ Hecke eigenforms of terms whose moduli are controlled by $\|G\|$, and the inequality is sharp only if their phases align, which the data say they never do. The growth in $N$ between $X = 2^{21}$ and $2^{23}$ is by a factor 1.0–1.2, as expected from Deligne's bound (the suprema are attained at $N$ with 16–96 divisors). We therefore expect:

*Conjecture* 5.1. For coprime $m, m'$ and $(N, mm') = 1$, $T(m, m'; N) = M(m, m'; N) + O_{\varepsilon}\big(N^{1/2+\varepsilon}\big)$ uniformly in $m, m'$. Consequently $r(n)r(N-n)$ has level of distribution $1/2$, and the bounds of Theorem 1.1 can be replaced by 8 and 5.

The conjecture is the exact analogue, for $r(n)$ and the sum orientation, of the uniformity in Topacogullari's theorem [11] for the divisor function, without the 7/64; it would follow from a

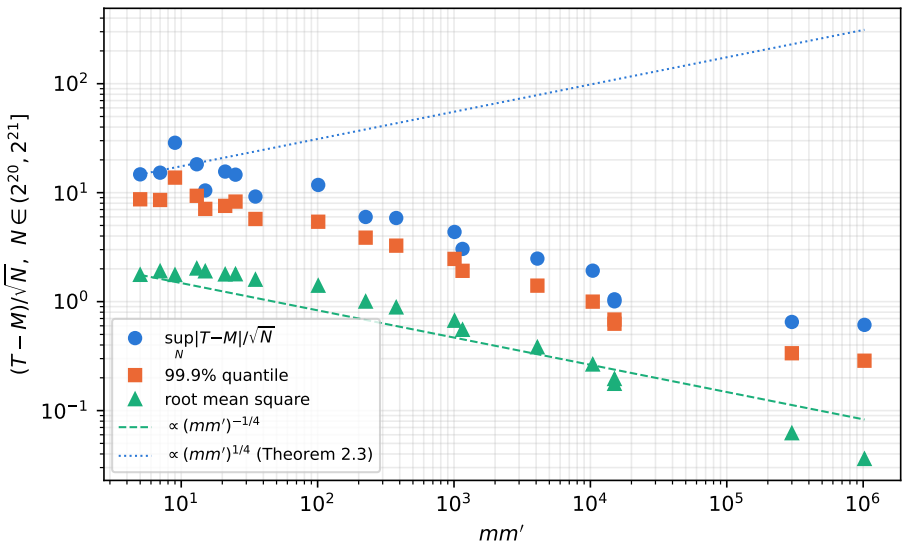



FIGURE 1. The data of Table 2 for $X = 2^{21}$ (all pairs computed), with the lines $(mm')^{-1/4}$ and $(mm')^{1/4}$ for comparison. The supremum does not grow with the level; the root mean square decreases like $(mm')^{-1/4}$.

uniform bound for the quantity $\sum_f \langle \Theta, f\rangle a_f(N)/\|f\|^2$ over a Hecke basis, i.e. from a first moment of the central values $L(1/2, f)L(1/2, f \otimes \chi)$ twisted by $\lambda_f(N)$ over the weight-2 forms of level $4mm'$.

## DATA AND CODE AVAILABILITY

The scripts behind Table 2 and Figure 1 (`sup_error_scan.py`), Table 1 (`certify_modular.py`), the choice of the sieve parameters (`modular_sieve_scan.py`) and the constant $c_h$ (`singular_constant.py`), together with their outputs, are available at

https://github.com/Ruqing1963/two-squares-theta-level

and archived at Zenodo, doi:10.5281/zenodo.23438288.

During the preparation of this work the author used the Claude language model and the `mpmath` Python library for the computer-assisted interval arithmetic verification of Section 4 and for the numerical experiments of Section 5. All analytical derivations and proofs were independently audited and verified by the author.

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